1 Planning (15 marks)

Defining the problem (3 marks)

P *f* is the independent variable or vary *f*. [1]

P I_0 is the dependent variable or measure I_0. [1]

P Keep V_0 constant. [1]

Methods of data collection (5 marks)

M Labelled/ workable circuit diagram of apparatus: including coil connected to power supply/signal generator. [1]

M Variable frequency ac power supply or signal generator (method of varying frequency). [1]

M Measure current using ammeter or p.d. across resistor. [1]

M Measure V across power supply or across coil e.g. voltmeter or c.r.o. explained. [1]

M Measure *f* using oscilloscope/read off signal generator. [1]

Method of analysis (2 marks)

A Plot a graph of $1/I_0^2$ against f^2 or V_0^2/I_0^2 against f^2 or equivalent.
Do not allow log-log graph. [1]

A $L = \sqrt{\frac{V_0^2 \text{gradient}}{4\pi^2}} = \frac{V_0}{2\pi}\sqrt{\text{gradient}}$ or $L = \sqrt{\frac{\text{gradient}}{4\pi^2}}$ [1]

Safety considerations (1 mark)

S Reasoned method to prevent overheating of coil or burns from coil. [1]

Additional detail (4 marks)

D Relevant points might include [4]
1 Use lower frequencies to produce larger currents
2 Keep resistance of circuit/coil constant
3 Additional detail on measuring *T* using timebase
4 *f* = 1 / period
5 Additional detail on measuring V_0 using *y*-gain
6 Relationship is valid if straight line, provided plotted graph is correct
7 Relationship is valid if straight line not passing through origin, provided plotted graph is correct (any quoted expression must be correct)
8 Detail on changing r.m.s. to peak

Do not allow vague computer methods. Ignore reference to iron core / other magnetic fields.

[Total: 15]

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